

Bluetooth for cyclists

Free2move has designed a bluetooth communications system for cyclists to enable easy communication

Charges for 3D glasses

Sony has decided to charge \$5-\$10 for 3D glasses for each blockbuster from May next year

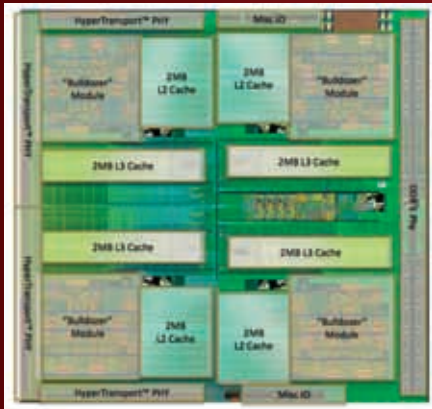
Bulldozer:

Understanding AMD's latest processor architecture

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All of AMD's processor architectures since AMD K7 have been pretty much the same. They were mostly revisions on the K7 architecture with minor modifications with each successive iteration. With Bulldozer, AMD designed a new architecture from scratch to incorporate a multitude of changes which it believes will help AMD regain some of the market share which it lost ever since Intel launched the Core 2 architecture in 2006.



Understanding the new Architecture On the outside

The Bulldozer processor looks pretty much like the Phenom/Deneb processor which used the AM3 941-pin socket. However, Bulldozer processors use 942 pins. This puts to rest rumours that AM3+ CPUs could be used with AM3 motherboards after getting a BIOS update. Unless you destroy one of the pins on your newly purchased processor, this is impossible. Motherboard manufacturers, however, have released new motherboards using older generation chipsets with the AM3+ socket. You can use Bulldozer CPUs on these motherboards after a BIOS update. A word of caution, AMD does not

support using Bulldozer processors on older chipsets.

Bulldozer has only one die with modules being locked permanently on models with less number of cores. So, a 4 core/2 module, a 6 core/3 module and the flagship 8 core/4 module Bulldozer are all the same die with certain cores being locked. Don't get your hopes up about getting a motherboard that can 'unlock' cores for your Bulldozer, AMD has ensured that none of the disabled cores will be usable.

The flagship processor, FX-8150 will come with liquid cooling from Asetek Solutions, it is also the OEM for Antec and Corsair's liquid coolers.



On the Inside

With the bulldozer architecture AMD has shifted from the multi-core approach to the module approach. The difference being that a module is AMD's answer to Intel's Hyper-Threading. After a lot of R&D, AMD came to the conclusion that desktop processors have to perform (FP) floating point calculations more as compared to integer calculations. So, instead of having two FP pipelines and 2 integer pipelines which would indicate two cores completely, bulldozer shares the Integer pipeline between two cores. Hence, each module contains 2 FP pipelines and 1 Integer pipeline, simulating two cores.

Bulldozer adds support for 3 new instruction sets; FMA4, CVT16 and XOP, all of which are useful for arithmetic calculations. It also supports Intel's AVX instruction set.

Finally, one can have 32 GB of RAM on their system, though there are very few who would need that kind of memory. Not to mention native support for 1866 MHz FSB. Though we are not sure, why AMD didn't opt for four channels of memory in the desktop processors which they have included in the Interlagos processors which are for servers.

The 990 FX Chipset

All FX models require the new AM3+ socket motherboards which come with the new 900-series chipsets.

The 900-series chipsets are also backwards compatible, so one can use the new AM3+ motherboards with older AM3 processors, if required. The only difference between the chipsets is the number of PCIE 2.0 lanes each supports. Moreover, SLI support is back with this chipset series which was absent since nForce 980a. RAID support remains the same with all chipsets supporting RAID 1, 0, 10. RAID 5 is supported on the 990FX chipset. There is support for native SATA 6 on these chipsets as well. The increased PCIE 2.0 bandwidth is something that AMD boards enjoy whereas Intel boards require the NF200 chip to match up in terms of bandwidth.

Courting controversies

Quite often new launches are mired in some controversy or the other and AMD was no stranger to this trend. First in line would be the launch dates. AMD never announced the dates but 'industry sources' first pegged the launch date to be June 2011. As June approached, the motherboard manufacturers released the respective 900-series motherboards, which in accordance to other processor